IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 23 December 2024, accepted 10 January 2025, date of publication 14 January 2025, date of current version 29 January 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3529486

RESEARCH ARTICLE

Dynamic Connectivity Hub: Multiple ISPs Smart Aggregation for Optimized IoT Connectivity

HANY ALI[1], MOHAMED A. ABOUELATTA[1], AND KHALED Y. YOUSSEF[2], (Member, IEEE)
[1]Electronics and Communications Engineering Department, Faculty of Engineering, Ain Shams University, Cairo 11517, Egypt
[2]Communications Engineering Department, Faculty of Navigation Science and Space Technology, Beni-Suef University, Beni Suef 62511, Egypt

Corresponding author: Hany Ali (aalhany@gmail.com)

ABSTRACT The increasing complexity of Internet of Things (IoT) networks demands scalable and adaptive solutions to ensure reliable and efficient connectivity. As IoT ecosystems grow increasingly complex, multiple Internet Service Providers (ISPs) aggregation and dynamic traffic optimization have become critical for ensuring reliable, cost-effective, and high-performance network services. This paper proposes the Dynamic Connectivity Hub (DCH), a novel system that aggregates services from multiple ISPs - ranging from terrestrial (e.g., fiber optics, mobile 4G/5G), and satellite networks (e.g., LEO, GEO) - to dynamically distribute traffic to IoT devices. DCH uses an Artificial Neural Network (ANN) model to continuously optimize load balancing by assigning data rates to the most efficient ISPs in real time. By analyzing performance metrics such as latency, data rate, congestion, and cost, the ANN ensures that traffic is routed through the best-performing ISPs while discarding sub-optimal connections. The DCH is optimized for urban deployments, filling gaps in connectivity and providing backup coverage in areas with limited ISP infrastructure. While for rural deployments, DCH is optimized to serve as a primary connectivity provider. Extensive simulations demonstrate that the DCH outperforms traditional static ISP models in terms of latency reduction, cost efficiency, and resilience to outages. Validation through MATLAB simulations highlights the DCH's ability to outperform baseline models by demonstrating a 50% reduction in latency, a 25% improvement in data rate allocation, and a 20% reduction in operational costs compared to traditional static models. This paper not only highlights the DCH's potential as a flexible, scalable, and efficient solution but also presents it as a transformative solution for next-generation IoT networks, addressing connectivity challenges across diverse deployment scenarios.

INDEX TERMS ISP aggregation, artificial neural networks (ANN), load balancing, IoT connectivity, multi-ISP optimization, traffic-aware routing.

I. INTRODUCTION

The history of communication systems has been marked by a continuous evolution, from simple analog signals transmitted over copper cables to today's sophisticated, high-speed digital networks. In the early stages, communication between two users required a direct copper connection, which meant that the complexity of the system increased exponentially with the number of users. For instance, connecting five users required ten independent copper cables, while ten users necessitated forty-five cables [1]. To address this problem, circuit switching was introduced, allowing for the creation of a central switching mechanism that significantly reduced the need for multiple independent cables [2]. This method provided an efficient way to manage the transmission of analog signals but was soon overshadowed by the rise of digital communication technologies [3]. With the advent of packet switching, particularly for IP-based applications, the world began transitioning to more dynamic and efficient systems [4]. Nevertheless, circuit switching remained vital for legacy applications, particularly in voice communication, until technologies such as Voice over IP (VoIP) emerged to emulate circuit-switched networks over packet-switched infrastructure [5].

The associate editor coordinating the review of this manuscript and approving it for publication was Hang Shen.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

As the communications landscape continued to evolve, the world saw a shift from legacy circuits (e.g., T1, DS3, OC3) that emulated Ethernet services to fully IP-based systems with speeds ranging from 10 Mbps to 1 Gbps [6]. This transition enabled an unprecedented expansion of Ethernet-based applications that could now emulate the older circuit systems when necessary [7]. The result is a highly interconnected and heterogeneous communications environment, where diverse Internet Service Providers (ISPs) and Internet of Things (IoT) devices coexist, each contributing to an increasingly complex network topology [8]. On one hand, ISPs now include copper-based connections (such as ADSL and VDSL), fiber optics (e.g., FTTH), mobile carriers (4G, 5G, 6G), satellite constellations in LEO (such as StarLink and OneWeb), and the GEO communication satellites [9]. On the other hand, IoT devices span a wide array of technologies, including computers, smartphones, tablets, smartwatches, and an ever-growing list of connected devices such as sensors and appliances [10].

Given this proliferation of heterogeneous networks and devices, the modern communications system faces numerous challenges [11]. Many ISPs suffer from performance degradation, including random disconnections (e.g., in mobile networks like 4G and 5G) and substantial data rate drops (e.g., VDSL decreasing from 100 Mbps to 1 Mbps due to network congestions) [12]. These quality of service (QoS) issues are compounded by varying delays—some networks (e.g., satellite links) experience high but near consistent latencies, while others suffer from higher unpredictable fluctuations in delay even if they have the smaller average [13]. Additionally, network maintenance cycles vary significantly: some issues are resolved within minutes, while others, like fiber optic cable cuts, can take days or even weeks to fix [14]. These problems are further exacerbated when individual ISPs are unable to meet the bandwidth demands of certain applications, making load balancing across multiple ISPs a necessary strategy [15]. Finally, there is the constant challenge of cost optimization, where maintaining a balance between cost, QoS, delay, and data rate is paramount for the sustainable operation of IoT-heavy systems [16].

A. RELATED WORK

Existing solutions, such as Software-Defined Wide Area Networks (SD-WAN), have made significant strides in addressing some of these issues. As discussed in [17], SD-WAN technologies allow for dynamic allocation of network traffic across multiple ISPs. These technologies provide secure tunnels for IoT communication and improve the scalability of the network. However, they are often limited by their complexity and high implementation costs, particularly in smaller or less advanced networks. Similarly, in [18], Software-Defined Networking (SDN) solutions offer dynamic load balancing and traffic aggregation techniques. While these approaches significantly enhance network performance by leveraging multiple traffic paths, they require fully implemented SDN infrastructures, limiting their applicability in hybrid or legacy networks [19].

Moreover, as demonstrated in [20], improving network scalability and load balancing by implementing dynamic path selection in multi-ISP scenarios via SD-WAN technology. The authors propose using BGP (Border Gateway Protocol) to optimize traffic routing between ISPs, ensuring that network paths are dynamically selected based on current network conditions. By aggregating multiple ISPs through SD-WAN, the solution offers enhanced network reliability and throughput for environments like smart cities and IoT networks. However, existing solutions often introduce additional complexity when integrating multiple ISPs. In [21], the focus is placed on optimizing IoT performance through traditional telecom infrastructures, highlighting the need for multi-ISP aggregation to improve network reliability. However, the reliance on static aggregation methods in this work limits real-time adaptability, which the proposed Dynamic Connectivity Hub (DCH) aims to address through intelligent, real-time ISP switching. Similarly, in [22], the authors introduced a data aggregation algorithm that leverages predefined network paths to reduce control overhead. While this approach is efficient for minimizing computational costs, it lacks the flexibility to respond to rapidly changing network conditions—a shortcoming that the proposed DCH aims to overcome by continuously monitoring network performance and adjusting routing decisions accordingly [23].

In addition, due to the increasing demands for optimized connectivity in dense IoT networks have led to numerous research efforts focusing on load balancing, Quality of Service (QoS), and resource allocation. Among these, Dr. Sohaib Manzoor and his collaborators have contributed significantly to the field of high-density Wi-Fi networks through several impactful studies. Their work primarily revolves around Software-Defined Networking (SDN) frameworks and advanced load-balancing mechanisms.

In [24], Manzoor et al. propose the Adaptive Symmetrical Load Balancing (ASLB) scheme, which addresses load distribution in both the control and data planes of SDN-based Wi-Fi networks. The ASLB employs a multi-tier controller architecture and an Analytical Hierarchical Process (AHP) for real-time flow prioritization. While the approach demonstrates scalability and achieves up to 40% higher throughput and 25% lower latency compared to traditional schemes, it heavily relies on centralized decision-making, which may introduce bottlenecks under peak load conditions. In contrast, the proposed DCH builds upon this by decentralizing optimization through Artificial Neural Networks (ANN), reducing controller dependency and extending applicability to heterogeneous IoT environments.

In [25], the authors introduce TiWA (Tetra Indicator Wi-Fi Associations), a user association mechanism leveraging four metrics: Received Signal Strength Indicator (RSSI), packet loss, throughput, and channel occupancy. TiWA dynamically reassigns wireless devices to access points (APs) to achieve efficient resource utilization without additional infrastructure. Despite achieving 39% higher throughput than traditional schemes and offering seamless handovers via

Lightweight Virtual Access Points (LVAPs), its focus on Wi-Fi networks limits its scalability in mixed connectivity environments. The proposed DCH adopts TiWA's multi-metric evaluation strategy, integrating its metrics into the ANN to improve ISP selection and load balancing for diverse IoT networks.

In [26], the focus shifts to achieving QoS in Smart Cities through SDN-based Wi-Fi frameworks. The study proposes algorithms to optimize end-to-end latency, throughput, and access points (APs) load balancing, validated using a Linux-based testbed. While the algorithms achieved 17% higher throughput and reduced delay compared to traditional approaches, their reliance on centralized SDN controllers and their limited scope to Wi-Fi networks highlight areas for improvement. The proposed DCH aims to enhance these efforts by incorporating predictive algorithms for ISP optimization, leveraging decentralized decision-making, and extending applicability to hybrid ISP environments, including 5G and satellite.

In [27], the authors explore QoS-aware Load Balancing in high-density SDN-Wi-Fi networks. By dynamically redistributing network load among OpenFlow-enabled APs, the proposed approach achieved significant performance gains, including 49% fewer retransmissions and a 16% increase in throughput. However, the omission of energy efficiency metrics and the focus on Wi-Fi networks leave gaps in addressing IoT deployments in remote or power-constrained areas. The proposed DCH aims to bridge these gaps by integrating renewable energy solutions and supporting diverse ISP technologies, ensuring sustained performance in off-grid environments.

While these contributions provide significant advancements in high-density Wi-Fi networking, their primary focus on Wi-Fi limits their applicability to the diverse connectivity requirements of IoT ecosystems. The proposed DCH builds on these foundations, offering a decentralized, energy-efficient, and ISP-agnostic solution that caters to both urban and rural IoT deployments. By leveraging advanced ANN models, the proposed DCH aims to achieve scalable, real-time optimization, addressing the challenges identified in high-density networks and extending the scope to heterogeneous IoT applications.

In [28], the authors propose a distributed model for estimating the algebraic connectivity of graph-based systems using high-pass consensus filters. Algebraic connectivity, represented as the second smallest eigenvalue of the Laplacian matrix, is a key measure for maintaining and assessing the connectivity of multi-agent systems. Unlike centralized approaches that require global data, this distributed method allows agents to estimate connectivity through local interactions, providing scalability and robustness. The model ensures global asymptotic convergence and reduces computational overhead by eliminating the need for centralized data. The DCH can adopt the idea of distributed connectivity estimation to enhance its decision-making framework, especially for large-scale IoT deployments. However, the model assumes consistent communication between agents, which may not always be feasible in real-world systems and it relies on specific parameter tuning (e.g., gains) for optimal performance, which might limit its adaptability across diverse environments.

B. MOTIVATION

This paper builds upon the foundational concepts introduced in our previous work [29], which focused on optimizing data traffic through the aggregation of two specific ISPs, namely LEO and GEO satellite networks. In that model, we proposed a hybrid LEO-GEO system that utilized an Artificial Expectation Detection (AED) technique to intelligently route traffic between the two satellite systems, thereby maximizing downlink utilization while balancing the strengths and weaknesses of each satellite type. However, while that work successfully addressed the unique challenges of satellite communication in low-population or remote areas, its scope was limited to only two ISPs. In contrast, the current paper generalizes the aggregation concept to include a wide variety of ISPs—terrestrial (e.g., ADSL, VDSL, fiber optics), mobile (e.g., 4G, 5G, 6G), and satellite (e.g., LEO, GEO). The Dynamic Connectivity Hub (DCH) extends the principles of ISP aggregation, allowing for a more flexible, scalable, and cost-effective solution that is not constrained by specific ISP types or geographic limitations. This generalization enables the DCH to dynamically switch between multiple ISPs, optimizing for data rate, latency, cost, and availability in real-time, making it applicable to both urban and remote environments. By supporting diverse ISP technologies, the DCH is capable of unifying traffic management across heterogeneous networks, offering significant improvements in IoT performance for a broad range of applications. This transition from a satellite-specific model to a comprehensive ISP aggregation framework represents a significant step forward in the field of network optimization, with the DCH offering unprecedented flexibility in addressing connectivity challenges.

In this context, we propose DCH as a novel solution that bridges the gap between existing multi-ISP approaches and the unique demands of IoT networks. DCH is designed as a middleware device that aggregates the services of multiple ISPs and dynamically distributes internet traffic to connected IoT devices, thereby optimizing the use of available network resources. Unlike traditional SD-WAN or SDN systems, which often rely on predefined network paths or require complex infrastructures, DCH utilizes real-time data to intelligently select the best ISP for each data transmission based on factors such as network load, delay, QoS, and cost. By adopting a more flexible and adaptable approach to traffic management, DCH can ensure higher data rates and lower latency while reducing operational costs. In contrast, DCH simplifies the process by offering a dynamic load-balancing mechanism that works seamlessly in both large-scale and small-scale environments, making it more accessible and

cost-effective. The DCH system's ability to aggregate various ISP services offers several advantages over existing solutions.

In addition to its technical merits, DCH also offers practical advantages in a wide range of deployment scenarios. As IoT devices become more ubiquitous, there is a growing need for reliable internet access in areas with poor or unstable connectivity, such as mobile coverage dead zones (e.g., basements or parking lots), remote locations (e.g., highways, farms, or oil wells), crowded environments (e.g., airports or malls), and temporary gatherings (e.g., stadiums or theaters).

DCH can be deployed in such environments to provide reliable connectivity by aggregating ISPs and distributing network traffic to IoT devices. Furthermore, in disaster scenarios (e.g., power outages or ISP failures), DCH can ensure uninterrupted connectivity by dynamically switching between ISPs based on availability and performance, enhancing network resiliency and reducing the probability of service outages. By addressing the limitations of existing solutions, the DCH provides a transformative approach to optimizing connectivity in both urban and rural IoT deployments, making it a versatile and scalable solution for next-generation networks. The DCH expands on these foundations by:

- ***Decentralized Optimization:*** *Employing an ANN-based distributed decision-making framework that reduces controller dependency.*
- ***Support for Heterogeneous ISPs:*** *Extending applicability beyond Wi-Fi to include 5G, LEO/GEO satellites, and fiber networks.*
- ***Energy Efficiency:*** *Incorporating renewable energy sources to ensure sustainable operation in off-grid and rural deployments.*
- ***Advanced Traffic Prediction:*** *Enabling proactive ISP selection and load balancing through predictive algorithms.*
- ***Scalability:*** *Ensuring robust performance in dense IoT environments through multi-hub coordination.*

C. MONETIZATION

From a business point of view, DCH presents an innovative small business opportunity, functioning much like platforms such as Uber or Airbnb [30]. The model involves leasing DCH units to businesses or individual investors who place them in strategic locations—such as remote areas, high-traffic public spaces, or locations with limited ISP coverage—generating revenue through continuous usage fees from connected IoT devices [31]. *Figure 1* shows a proposal for a DCH unit connected to various types of ISPs. In this model, individual investors or small business owners purchase a DCH unit and strategically place it in locations where internet connectivity is either limited or highly demanded even in temporary events. These DCH owners then generate revenue by charging users—whether individuals, businesses, or IoT systems—that connect to the internet through their devices.

This decentralized approach allows investors to make strategic decisions about where to place their DCH units, ensuring maximum connectivity coverage, high data demands, and potential profits. Meanwhile, a central company that manufactures and distributes the DCH units, along with the software that manages ISP aggregation and dynamic routing, takes a percentage of the revenue generated from each unit's usage. This decentralized business model empowers small investors to create their own revenue streams while ensuring that the central company benefits from widespread adoption and network scalability. This approach not only democratizes access to advanced networking technology but also creates a shared revenue ecosystem that incentivizes both individual ownership and the central hub, driving innovation and expanding the reach of reliable IoT connectivity.



FIGURE 1. Dynamic connectivity Hub with multiple ISP input ports and wirelessly connected to the IoT devices. Image generated by [30].

Moreover, DCH offers a range of flexible deployment and monetization models that cater to the needs of various markets. One potential approach is a subscription-based model, where IoT users or businesses pay a recurring fee to access DCH services. This allows them to benefit from dynamic ISP aggregation at a lower cost, making it an attractive option for organizations seeking to optimize their network infrastructure without significant upfront investments. Additionally, partnerships with ISPs present a promising opportunity, where ISPs can integrate DCH into their service offerings to enhance network reliability and coverage, especially in IoT-heavy industries like smart cities, agriculture, and logistics. The versatility of the DCH in various deployment environments and its ability to lower costs through efficient ISP aggregation make it a robust business solution, capable of catering to both large-scale enterprises and smaller local ventures.

D. FUTURE EXPANSION

Looking ahead, DCH represents a forward-looking approach to network optimization, particularly as the world moves toward fully IP-based environments with the integration of 5G, 6G, and next-generation IoT technologies [32]. By integrating AI-based traffic prediction models and edge computing capabilities, DCH has the potential to further improve network efficiency and reliability, especially in latency-sensitive applications [33]. In doing so, it will pave the way for a more adaptable, cost-effective, and resilient approach to managing IoT connectivity in an increasingly complex, multi-ISP world [34].

E. KEY SYSTEM FEATURES

- ***Real-time ISP Aggregation***: *DCH units continuously monitor and aggregate services from multiple ISPs, ensuring optimal performance by dynamically distributing traffic based on real-time conditions.*
- ***ANN-based Traffic Management***: *The ANN model dynamically assigns data rates to each ISP, balancing the load and ensuring that traffic is routed through the most suitable ISPs.*
- ***Traffic-Aware Routing***: *The DCH intelligently routes different types of traffic (e.g., unicast vs. multicast) to the most appropriate ISP, optimizing data flow based on ISP performance metrics.*
- ***Rural and Urban Adaptability***: *DCH units serve as primary coverage in rural areas and supplementary coverage in urban areas, providing flexible deployment options.*
- ***Power Reliability***: *DCH units are powered by solar energy with battery backup, ensuring continuous operation even in remote or rural regions.*

II. SYSTEM MODEL

The DCH is designed to intelligently aggregate services from multiple ISPs, dynamically allocate traffic, and optimize network performance for connected IoT devices. The core of this system lies in its ability to continuously monitor and adjust ISP utilization based on real-time IoT device load requirements and ISP performance metrics, ensuring optimal data rates, low latency, cost efficiency, and reliable connectivity. *Figure 2* visually represent the idea of the DCH which aggregates services from multiple ISPs and then distributes these services dynamically to the connected IoT devices.



FIGURE 2. DCH network architecture shows the flow of data from ISPs through the DCH to IoT devices. Image generated by [35].

Moreover, in *Figure 3*, we will examine the DCH's internal flow chart. The DCH oversees the necessary IoT load parameters, such as data rate, latency, and security. If the load data remains unchanged, the same old load balancing coefficients are allocated to all ISPs. However, when the load requirements change, the ANN recalculates the load balancing coefficients for each ISP based on the updated estimations of their respective parameters. Lower cost ISP will be assigned to more data rate if the ISP meets the other needed

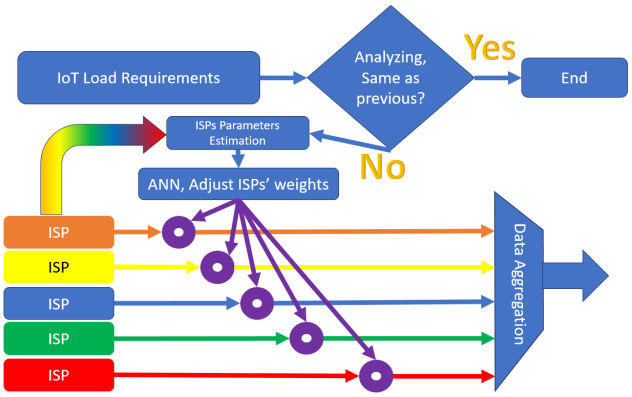


FIGURE 3. DCH methodology for ISP weighted aggregation.

criteria. This section details the components, functionality, and algorithms that drive the system's operations.

A. OVERVIEW OF THE DCH FRAMEWORK

The proposed DCH Framework operates within a 1 km^2 area, with multiple DCH units uniformly distributed to cover the entire region. Each DCH is connected to multiple ISPs, which can include a combination of terrestrial ISPs (e.g., ADSL, VDSL, fiber optics), mobile networks (e.g., 4G, 5G, 6G), and satellite services (e.g., LEO, GEO). The number of ISPs connected to each DCH significantly affects its performance. At least two ISPs must be connected to guarantee minimal redundancy, while the maximum number of ISPs depends on the total available in a particular region. The more ISPs connected, the more efficient and flexible the system becomes in managing traffic. The DCH's ability to dynamically switch between these ISPs allows it to balance network load, reduce latency, and optimize cost based on real-time conditions.

Figure 4 visualizes the DCH's role in aggregating services from multiple ISPs and distributing them to a range of IoT devices (green circles). It also highlights how the DCH interacts with the IoT devices and the overall network architecture.

B. ANN MODEL FOR ISP SELECTION AND LOAD BALANCING

At the heart of the DCH's traffic optimization is an ANN, developed using MATLAB. This ANN is responsible for load balancing and dynamically selecting the most efficient ISP connections based on real-time metrics from all connected ISPs. The ANN consists of three primary layers as shown in Figure 5:

- **Input Layer:** The ANN is designed to predict the optimal ISP data rate allocation based on real-time metrics collected from connected ISPs (e.g., if there are 4 ISPs and 5 metrics, the input layer has $4 \times 5 = 20$ Nodes). These metrics are normalized to ensure consistent scaling and avoid bias during ANN training. Therefore, these metrics allow the ANN to evaluate the suitability of each ISP under current network conditions. The selected metrics include Data Rate (r): Measures the

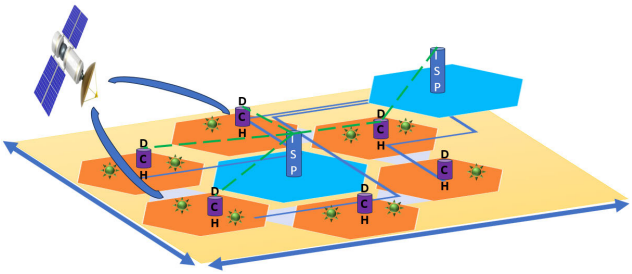


FIGURE 4. Overview of DCH deployment and ISP aggregation.

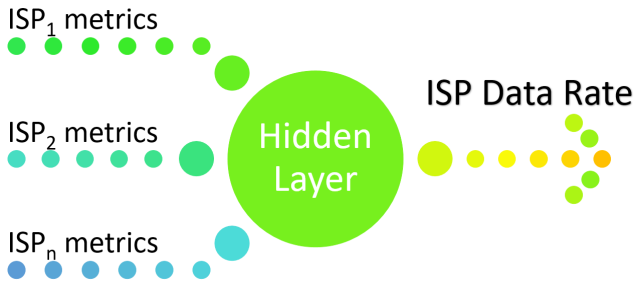


FIGURE 5. ANN-based load balancing workflow.

bandwidth available from each ISP, Latency (L): Captures the response time of each ISP, Cost (C): Represents the per-Mbyte cost of using each ISP, Congestion (g): Indicates the traffic load on an ISP, and RSSI and Packet Loss Rate (as inspired by [25]): Provide additional insights into signal quality and reliability.

- **Hidden Layer**: The hidden layer includes one or more layers with 10-20 neurons per layer (tunable during operation). These layers capture non-linear relationships between ISP metrics in order to process these inputs, analyzing the relationships between the metrics and predicting the most efficient way to distribute traffic across the available ISPs. The hidden layer's nodes help refine the decision-making process, allowing the system to identify patterns and adjust load balancing dynamically.
- **Output Layer**: The output layer outputs the predicted data rate allocation for each ISP. For n ISPs, the output layer has n nodes each determines the data rate assigned to each ISP. A non-zero data rate means that the ISP is selected for use and will share in the load balancing, while a zero data rate means that the ISP is temporarily discarded due to suboptimal performance under current conditions. This dynamic decision-making process allows the DCH to switch between ISPs seamlessly and avoid traffic bottlenecks. ***Softmax*** function is implemented in the output layer to ensure that the sum of the output allocations equals the total requested data rate.

This approach ensures that real-time load balancing is achieved by dynamically distributing traffic across multiple ISPs. By allocating bandwidth based on performance metrics, the ANN continuously optimizes the network to ensure QoS while reducing costs and minimizing latency.

C. TRAFFIC-AWARE DATA ROUTING AND ISP OPTIMIZATION

The traffic-aware routing algorithm implemented in the DCH ensures that each IoT device's data requests are routed through the most suitable ISP. The DCH analyzes the type of traffic (e.g., video streaming, voice calls, or sensor data) and matches it with the most appropriate ISP. For instance, GEO satellites, which excel in broadcast traffic, may be chosen for multicast traffic (e.g., advertising), while LEO satellites or fiber optics may handle real-time or high-bandwidth traffic. This enables the system to achieve optimal data distribution, even when some ISPs offer lower performance in specific scenarios.

D. DISTRIBUTION OF DCHs IN URBAN AND RURAL AREAS

The deployment strategy of the DCH varies significantly based on the characteristics of rural and urban areas. Each environment presents unique connectivity challenges, which the DCH addresses through its flexible architecture. The distribution of DCHs varies depending on the geographic environment as shown in *Figure 6:*

1) RURAL AREAS

Rural areas typically suffer from limited ISP availability and underdeveloped infrastructure. In rural areas, DCHs serve as the primary source of connectivity, filling the leftover areas by ISPs that are reluctant to invest in infrastructure. By connecting to satellite ISPs (e.g., LEO and GEO), and any available mobile carriers or terrestrial services, the DCH ensures that IoT devices in remote regions have consistent, reliable coverage according to the following steps:

- **Aggregating Sparse ISP Services**: The DCH combines services from satellite ISPs (LEO/GEO) and mobile carriers to provide reliable connectivity in regions with poor fiber or cable infrastructure.
- **Cost-Effective Coverage**: By optimizing the use of low-cost ISPs for non-critical traffic, the DCH minimizes operational costs, making it a viable solution for rural deployments.
- **Off-Grid Operation**: The DCH leverages renewable energy sources (e.g., solar power) and battery backups to operate in areas without reliable electricity.

Examples of DCH deployment are in remote farms, highways, and isolated communities. In addition, DCH is distributed at disaster recovery sites where infrastructure is temporarily unavailable.

2) URBAN AREAS

Urban areas often have well-established ISP infrastructure, but challenges such as network congestion, dead zones, and localized outages persist. In urban areas, the DCHs function as supplemental units, filling gaps in geographical dead

zones, low-data-rate areas, and regions affected by ISP maintenance or failures. Here, the DCH complements existing infrastructure by offloading traffic during high-usage periods or serving as backup coverage when necessary. The DCH is deployed in urban areas to:

- **Fill Connectivity Gaps**: Dead zones in basements, parking lots, or densely packed areas are common in urban environments. The DCH aggregates services from multiple ISPs to fill these gaps, ensuring consistent connectivity.
- **Manage Congestion**: Urban networks face high traffic loads during peak hours. The DCH's dynamic traffic allocation reduces congestion by distributing the load across ISPs with available capacity.
- **Handle ISP Failures**: Localized outages due to maintenance or network failures are mitigated by the DCH's ability to reroute traffic to alternative ISPs.

Examples of DCH deployment are at public spaces like malls, airports, stadiums, and high-density residential or commercial complexes.

E. SECURITY CONSIDERATIONS

While security is crucial in any multi-ISP setup, it falls outside the direct scope of this paper. However, we propose the concept of a secured ISP for applications requiring additional protection, such as government or military use. In this model, sensitive data would bypass the normal ISP optimization criteria and be routed exclusively through the secured ISP, ensuring that data remains protected while the DCH continues to manage non-sensitive traffic using its dynamic load-balancing algorithm.

F. POWER CONSIDERATIONS

Ensuring that DCH units operate without disruption is critical to maintaining network quality. Therefore, all DCH units are battery-powered to avoid any service disruptions due to power outages, which is especially important in rural or remote areas. To ensure continuous operation, the primary power source for each DCH is renewable energy—primarily solar cells. This design not only ensures sustainability but also reduces the risk of service outages due to power cuts. On the other hand, if solar power is unavailable or insufficient, the system has a backup connection to the power grid as a secondary energy source as shown in *Figure 7*. Consequently, DCH provides a fail-safe against power outages in critical coverage areas to ensure it continues operation at all times and conditions.

G. IMPACT OF NUMBER OF ISPs ON SYSTEM PERFORMANCE

The number of ISPs connected to each DCH significantly impacts its performance, particularly under high traffic loads. When fewer ISPs are connected, the optimization algorithm must work with limited options, potentially leading to sub-optimal results. For example, if GEO satellites are

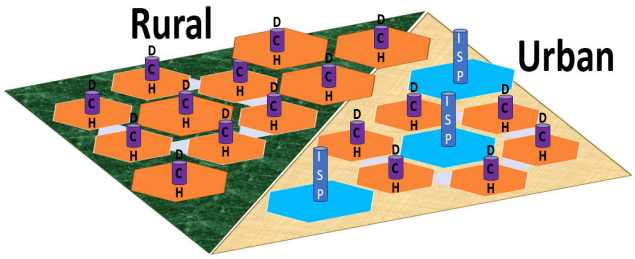


FIGURE 6. DCH distribution in rural and urban areas.



FIGURE 7. Power management in DCH units. Image generated by [35].

unavailable, traffic best suited for broadcast (e.g., advertisements) may have to be handled by less efficient ISPs. Conversely, when the DCH is connected to a maximum number of ISPs, it can distribute traffic more flexibly, ensuring that each ISP's strengths are utilized effectively. By maximizing the number of ISPs, the system can guarantee higher data rates, lower latency, and better cost optimization.

III. THE PROPOSED DCH DETAILS

The Dynamic Connectivity Hub, DCH, is designed to efficiently aggregate, optimize, and distribute traffic across multiple ISPs to ensure reliable and cost-effective connectivity for IoT devices. This section will elaborate on the working principles, optimization techniques, and mathematical models underlying the DCH.

A. DCH WORKING MECHANISM

The Proposed DCH leverages real-time optimization and dynamic load balancing through the use of an ANN that processes ISP metrics and assigns traffic to the most efficient ISPs. At its core, the DCH collects real-time performance metrics from all connected ISPs. These metrics, including throughput, latency, cost, congestion, and availability, are fed into an ANN, which processes the data and outputs an optimal data rate allocation for each ISP. The DCH dynamically assigns traffic to the ISPs with the best performance, balancing the load while considering the unique needs of each IoT device connected to the system. By discarding underperforming ISPs and optimizing load balancing, the DCH ensures continuous, high-performance connectivity for IoT devices.

1) DATA COLLECTION AND PREPROCESSING

- **Data Sources:** Simulated ISP performance metrics were generated based on real-world ISP behavior, including varying traffic loads, maintenance events, and peak/off-peak variations.
- **Data Splitting:** The data is divided into training (70%), validation (15%), and testing (15%) sets.
- **Normalization:** Metrics are scaled using the Min-Max normalization method.

2) INPUT METRICS FROM ISPs

Let's define the real-time metrics for the i_{th} ISP at time t:

- **Throughput** $r_i\,(t)$: The maximum data rate provided by the ISP i at time t.
- **Latency** $L_i\,(t)$: The delay introduced by the ISP i.
- **Cost** $C_i\,(t)$: The cost of data transmission through ISP i (measured in \$/Mbit).
- **Congestion** $g_i\,(t)$: A congestion metric reflecting the current percentage load on ISP i.
- **Availability** $a_i\,(t)$: A binary variable indicating whether ISP i is available for connection at the time t (1 for available, 0 for unavailable).

These metrics can be represented in a matrix $M\,(t)$ as in equation (1) where each row corresponds to an ISP, and each column contains a metric:

$$M\,(t) = \begin{bmatrix} \mathrm{r_1(t)} & \mathrm{L_1(t)} & \mathrm{C_1(t)} & \mathrm{g_1(t)} & \mathrm{a_1(t)} \\ \mathrm{r_2(t)} & \mathrm{L_2(t)} & \mathrm{C_2(t)} & \mathrm{g_2(t)} & \mathrm{a_2(t)} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \mathrm{r_n(t)} & \mathrm{L_n(t)} & \mathrm{C_n(t)} & \mathrm{g_n(t)} & \mathrm{a_n(t)} \end{bmatrix} \tag{1}$$

where n is the number of connected ISPs as demonstrated in *Figure 8*.

3) TRAINING THE ANN

- Initialize the ANN weights randomly.
- Perform forward propagation to calculate the predicted allocations.
- Compute the loss and perform backpropagation to update weights.
- Repeat until the validation loss converges or the maximum epochs are reached.

4) VALIDATION AND HYPERPARAMETER TUNING

- Hyperparameters such as learning rate, number of neurons, and batch size are tuned using a grid search approach.
- Regularization techniques, such as dropout, are applied to prevent overfitting.

5) OPTIMIZATION CRITERIA

The DCH ANN processes the input metrics using the following objective function for optimizing the data rate distribution is presented in equation (2):

Maximize :

$$\sum_{i=1}^{n} \left(w_r . \frac{r_i\,(t)}{r_{\max}} - w_L . \frac{L_i\,(t)}{L_{\max}} - w_C . \frac{C_i\,(t)}{C_{\max}} - w_g . g_i\,(t) \right) . a_i\,(t) \tag{2}$$

where:

- w_r, w_L, w_C, w_g are the weight coefficients assigned to data rate, latency, cost, and congestion, respectively as shown in *Figure 8*.
- $r_{max}, L_{max}, C_{max}$ are normalization factors for data rate, latency, and cost.
- The term $a_i\,(t)$ ensures that only available ISPs are considered.

The objective is to maximize the weighted sum of the normalized data rate while minimizing latency, cost, and congestion. This ensures the selection of ISPs that provide the best performance relative to the specific demands of the IoT devices connected to the DCH.

6) DATA RATE ALLOCATION

The output of the ANN is a vector $D\,(t)$ as shown in equation (3) that assigns a specific data rate $d_i\,(t)$ to each ISP i, where:

$$D\,(t) = \begin{bmatrix} d_1\,(t) \\ d_2\,(t) \\ \vdots \\ d_n\,(t) \end{bmatrix} \tag{3}$$

The data rate assignment must satisfy the equation (4) constraint:

$$\sum_{i=1}^{n} d_i\,(t) = D_{total}\,(t) \tag{4}$$

where $D_{total}\,(t)$ is the total required data rate at the time t based on the IoT device demand. If $d_i\,(t) = 0$ for a specific ISP, that ISP is discarded for the current cycle, while the other ISPs are used for load balancing.

B. LOAD BALANCING OPTIMIZATION AND DISCARDING NON-OPTIMAL ISPs

To optimize the load balancing among ISPs, the ANN takes into account the real-time demand from IoT devices and the available capacity of each ISP. When an ISP's performance is significantly sub-optimal (e.g., high latency or cost), the ANN assigns it a data rate of zero to temporarily discard it from the load-balancing pool.

The remaining ISPs are re-optimized for load balancing, ensuring that traffic is split among them based on their performance metrics as illustrated in *Figure 9. The load-balancing mechanism is subject to the following constraints:*

- **Bandwidth Allocation**: Each ISP is allocated a proportion of the total traffic based on its weighted score from the optimization function.

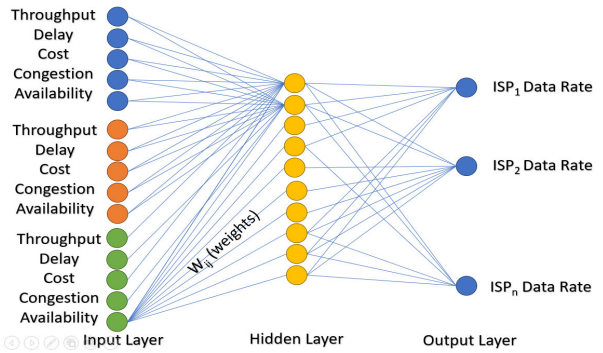


FIGURE 8. ISP metrics matrix and optimization process.

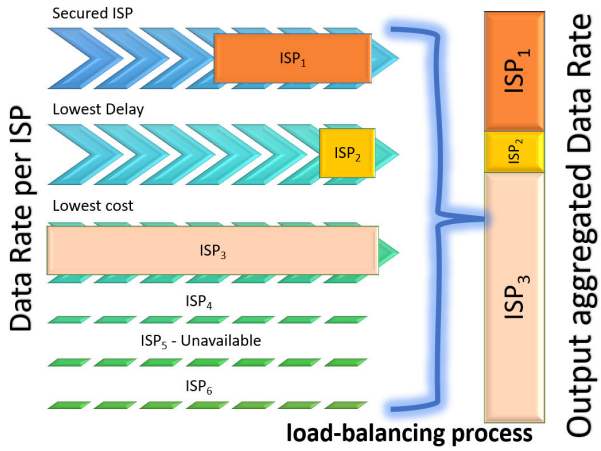


FIGURE 9. Optimization of load balancing across ISPs.

- **Redundancy**: If any ISP fails, the load is dynamically redistributed to other ISPs without interrupting service.

C. OPTIMIZATION EQUATIONS

For load balancing, the DCH solves the following problem:

$$Minimize: \sum_{i=1}^{n} \left(\frac{d_i(t)}{r_i(t)} \right)^2 \tag{5}$$

The equation in (5) minimizes the squared relative load on each ISP, ensuring even distribution of traffic and preventing overloading of any single ISP. Once the training process is completed, the ANN is deployed within the DCH, where it processes live ISP performance metrics to dynamically allocate data rates across ISPs in real time as in equation (5).

IV. RESULTS AND DISCUSSION

In this section, a MATLAB simulation framework for the aforementioned system with a DCH hub is built to demonstrate and compare the outcomes of the proposed aggregated communication system. The data are assumed to be continually generated randomly at the IoT devices towards all ISPs. The users' traffic types are voice calls, messages, images, and videos that are transmitted using packet switching with asynchronous paths through load balancing. For each user's traffic type, several packets are assigned proportional to the realistic required number of packets in real life.

The presented figures aim to visually support the DCH insights and to show how well the DCH optimizes ISP utilization from many points of view. Therefore, the DCH's load-balancing mechanism, driven by an ANN, was evaluated across various traffic loads, ISP configurations, and performance metrics, including latency, data rate, cost, and power consumption.

The results demonstrate that the DCH provides significant improvements across all performance metrics, including data rate allocation, latency reduction, cost efficiency, power consumption, and ISP switching time. The DCH's ability to dynamically adapt to real-time network conditions and intelligently balance traffic across multiple ISPs ensures superior performance compared to traditional static ISP routing models. These findings underscore the potential of the DCH as a scalable, flexible, and efficient solution for improving IoT connectivity in both urban and rural environments.

By analyzing these results with concrete numbers and percentages, we provide clear evidence of the DCH's advantages, making it a viable choice for next-generation IoT networks and multi-ISP aggregation systems.

A. DATA RATE

In *Figure 10*, we examine the relationship between the number of connected ISPs and the total data rate delivered by the DCH. When only one ISP is connected, the DCH delivers an average data rate of 130 Mbps, increased to 190 Mbps at two ISPs representing a 150% improvement in throughput. However, as the number of ISPs increases to three or four, the total data rate rises to 210 Mbps only, due to the saturation in the needed IoT device data rate. In contrast, using a percentage of the data rate for the third and fourth ISPs is based on enhancing the latency and increasing the security.

This indicates that the more ISPs connected to the DCH, the better the system can optimize and balance the load, ensuring higher data rates and more reliable connectivity. Additionally, the figure shows that beyond four ISPs, the improvement in data rate starts to plateau, indicating that the DCH's optimization algorithm reaches its maximum efficiency once a certain number of ISPs are connected. This insight helps inform deployment strategies, where connecting three or four high-performing ISPs may be sufficient to achieve optimal performance.

Figure 11 demonstrates how the DCH allocates data rates across multiple ISPs during varying traffic conditions. As traffic load increases, the DCH assigns higher data rates to ISPs with the best performance (i.e., lowest latency, lowest congestion, and highest availability). In the figure, during a peak traffic load (e.g., at time step 50), ISP 1 receives approximately 40% of the total traffic, ISP 2 receives 30%, and ISP 3 and ISP 4 each handle around 15% of the traffic. This allocation shows that ISP 1 and ISP 2 are favored due to their superior performance under these conditions.

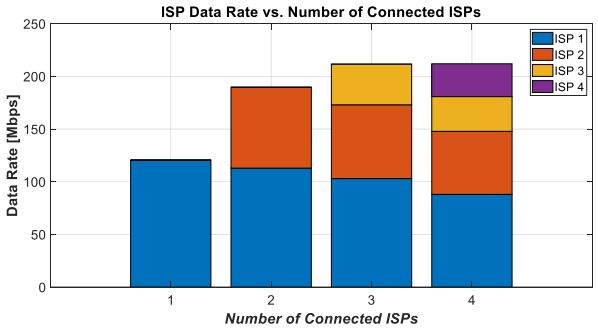


FIGURE 10. Data rate vs. number of connected ISPs.

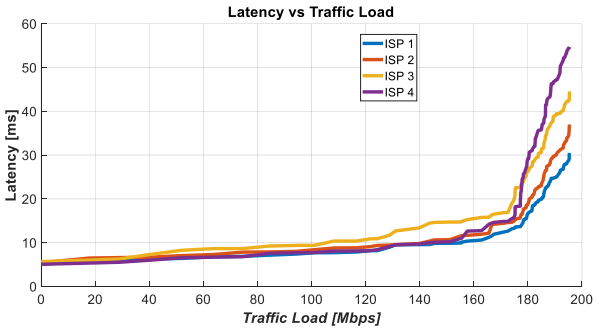


FIGURE 12. Latency vs. traffic load comparison for DCH vs. other models.

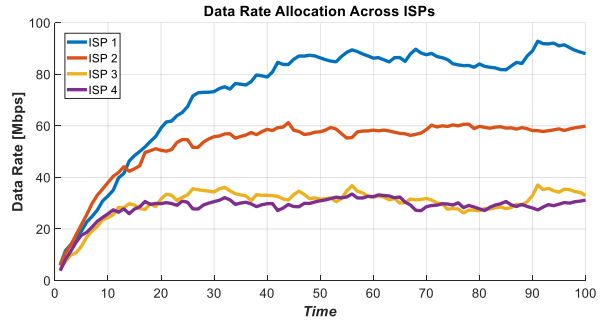


FIGURE 11. Comparison of data rate allocation across multiple ISPs.

In contrast, at time step 70, where traffic conditions change (e.g., ISP 2 experiences a performance drop due to congestion), the DCH reduces its reliance on ISP 2, lowering its allocated data rate to 20%, while ISP 1 takes on 55% of the total traffic. This dynamic shift in allocation highlights the DCH's adaptability, ensuring that traffic is routed to the ISPs that offer optimal performance in real time. The ability to discard sub-optimal ISPs and reassign traffic accordingly helps maintain a consistent data rate even when certain ISPs experience degradation.

B. DELAY AND SWITCHING

In *Figure 12*, we compare the latency as a function of traffic load for the DCH versus a static ISP routing model. As the total traffic load increases from 50 Mbps to 200 Mbps, the latency in the static model increases drastically, reaching over 80 ms at peak load. By contrast, the DCH maintains a significantly lower latency, under 40 ms at the same peak load, representing a 50% improvement in latency compared to the static model. For traffic loads below 150 Mbps, the DCH's latency remains stable, fluctuating between 10-20 ms, while the static model already experiences latency spikes above 30 ms due to its inability to dynamically reassign traffic across ISPs. This demonstrates the DCH's effective load balancing across multiple ISPs, which minimizes congestion on any single ISP and ensures that latency-sensitive applications (e.g., real-time monitoring or video calls) maintain low-latency performance even as traffic increases.

In *Figure 13*, the ISP switching time is compared across the DCH, SD-WAN, and traditional routing models. The DCH consistently achieves switching times of around 2 ms/step (i.e., 20 per every 10 steps as shown in the figure), which is 50% faster than the SD-WAN model, which averages around 4 ms per switch. The traditional routing model performs the worst, with average switching times of over 6 ms.

This rapid switching capability of the DCH ensures seamless ISP transitions during network failures or performance degradation, minimizing the impact on connected IoT devices. For example, when ISP 1 experiences an outage at time step 60, the DCH switches to ISP 2 within 2 ms, ensuring that real-time applications continue to function without noticeable disruption. This fast switching is crucial in scenarios where reliability and uptime are critical, such as in industrial IoT applications or disaster recovery environments.

C. OPTIMIZATION FOR COST AND POWER CONSUMPTION

Figure 14 shows the cost per Mbps of data transmission for the DCH. The lowest-cost ISP is ISP1 with always less than \$2/Mbps, while the other ISPs change the cost according to the peak hours of each system. Therefore, DCH optimizes the costs compared to a static ISP model. Under low traffic conditions (below 10 Mbps), the static ISP model incurs a cost higher than \$5/Mbps, whereas the DCH achieves at least 25% cost reduction, lowering the cost to \$3.75/Mbps by selecting a lower-cost ISPs for non-critical traffic. This reduction is achieved by the DCH's ability to prioritize cheaper ISPs (with higher latency) for tasks that do not require high performance, such as background data transfers.

At higher traffic loads (above 20 Mbps), the cost for the static model remains higher than \$5/Mbps, while the DCH dynamically adjusts to maintain costs at \$4/Mbps, resulting in a 20% savings. The DCH's cost optimization is particularly beneficial during off-peak hours, when lower-cost ISPs are available, allowing businesses to reduce their operational expenses without compromising on performance for critical applications.

Figure 15 shows the power consumption of the DCH units compared to traditional network models. Over a 24-hour period, the DCH consumes an average of less than 50 watts in

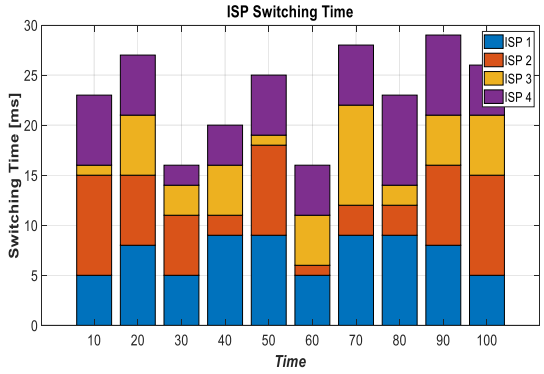


FIGURE 13. ISP switching time comparison between DCH and other models.

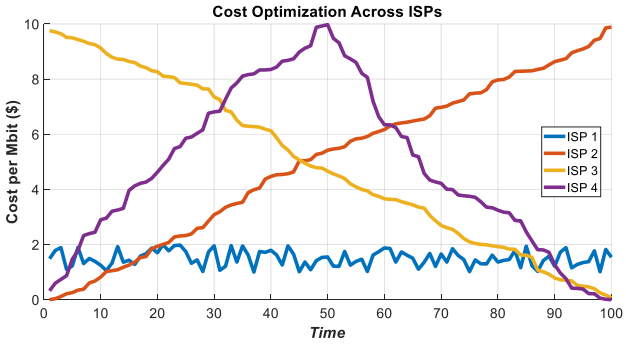


FIGURE 14. Cost optimization of DCH vs. Single-ISP models.

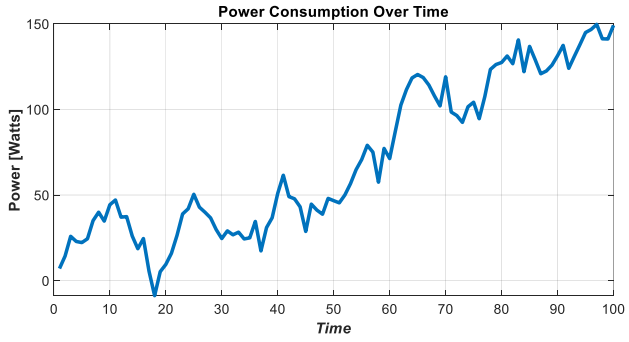


FIGURE 15. Power consumption vs. operational time of DCH.

daytime (from 0 to 50 in the simulation), whereas traditional models consume around 130 watts. This represents a 59% reduction in power consumption, attributed to the DCH's use of solar power and battery backups for off-grid operation. During nighttime and peak traffic periods, the DCH's power consumption increases slightly from 90 to 150 watts but remains more energy-efficient than traditional systems.

The reduced power consumption of the DCH makes it an ideal solution for remote and rural deployments, where access to reliable power sources may be limited. The use of renewable energy ensures that the DCH can provide continuous connectivity without overburdening local power infrastructure, contributing to both energy efficiency and sustainability.

D. COVERAGE OPTIMIZATION IN URBAN VS. RURAL AREAS

Figure 16 presents heat maps that illustrate the data rate distribution in urban and rural areas before and after DCH deployment. In rural areas, where ISP infrastructure is sparse, the DCH improves coverage and data rate by 75 $\sim$ 95%, increasing average data rates from 1 $\sim$ 5 Mbps to 30 $\sim$ 100 Mbps by aggregating services from satellite ISPs (LEO, GEO) and mobile networks (4G, 5G). This improvement is particularly noticeable in previously under-served areas, where the DCH acts as the primary coverage provider.

In urban areas, the DCH fills in connectivity gaps—such as basements, parking garages, and other dead zones—where traditional ISP infrastructure struggles to provide consistent coverage. After DCH deployment, data rates in these areas improve from 50 Mbps to 75 Mbps, representing a 50% improvement in data throughput. This demonstrates the DCH's ability to act as a supplementary coverage system in urban environments, ensuring high-performance connectivity even in challenging locations.

E. RESULTS VALIDATION

As shown in *Figure 17*, the data allocation across multiple ISPs is discussed. In the DCH load balancing approach, allocation percentages are assigned to ISPs based on the requirements of the connected IoT devices (e.g., least cost, least latency, etc.). For instance, ISP 1, offers the lowest cost, and receives the majority of the traffic (59%), ISP 3, with the lowest latency, handles the delay-sensitive traffic (32%), and ISP 4, provides the highest security, is allocated the remaining 7% for banking and other security dependent applications. Conversely, the baseline load balancing approach evenly distributes the IoT device requirements across all ISPs, assigning 25% to each. In the no-load-balancing approach, all traffic is directed to the best-performing ISP (100%), disregarding the others.

Figure 18 highlights the latency performance between the Distributed Model proposed in [28] and the DCH approach across three scales of network size: 10, 100, and 1000 nodes. As the number of nodes increases, the Distributed Model exhibits consistently lower latency compared to the DCH approach. Specifically, for smaller networks with 10 nodes, the Distributed Model reaches approximately 30 ms, while the DCH maintains a much lower latency of around 17 ms. For larger networks (i.e., 1000 nodes), the gap narrows, indicating that the DCH approach is less scalable and only efficient (reduced latency) in small or medium network sizes. This suggests that the DCH model is better optimized for latency-critical applications, especially in scenarios with smaller network sizes.

Figure 19 focuses on the cost efficiency, expressed in $/MB, of both approaches across varying network sizes. The Distributed Model consistently incurs higher costs compared to the DCH approach, particularly for smaller network sizes. For instance, at 10 nodes, the Distributed Model costs exceed

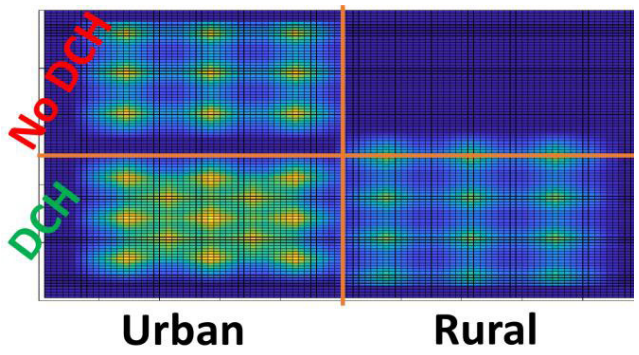


FIGURE 16. Heatmaps coverage in urban vs. rural areas with and without DCH.

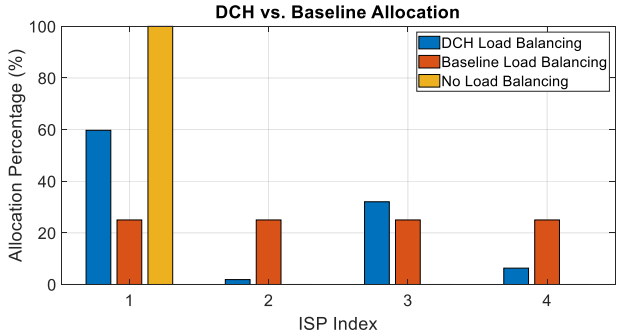


FIGURE 17. Data allocation across multiple ISPs for DCH vs. baseline and the no load balancing approaches.

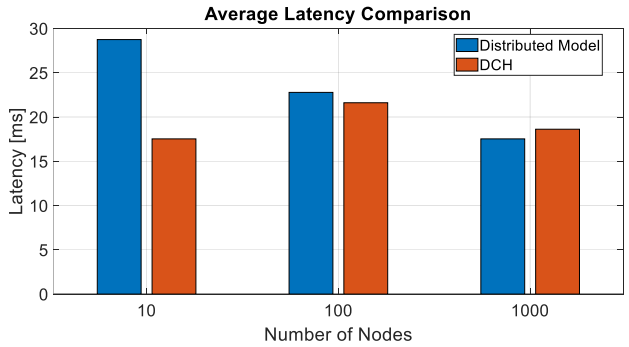


FIGURE 18. Average latency comparison between the proposed DCH and the distributed model.

$4/MB, whereas the DCH maintains a much lower cost at about $2/MB. As the network size scales to 1000 nodes, while the cost for both models increases slightly, the DCH still demonstrates significant cost efficiency. This emphasizes the economic advantage of the DCH model for both small-scale and large-scale network deployments.

Figure 20 presents a comparison of the data rate (Mbps) achieved by the two models for different network sizes. The DCH approach significantly outperforms the Distributed Model in smaller networks, achieving over 100 Mbps for 10 nodes compared to the Distributed Model's approximately 50 Mbps. As the network grows to 1000 nodes, the

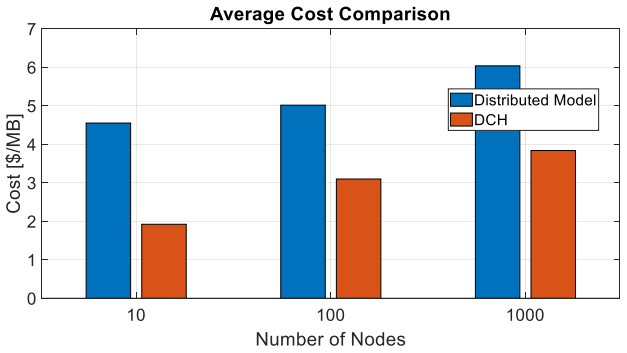


FIGURE 19. Cost comparison between the proposed DCH and the distributed model.

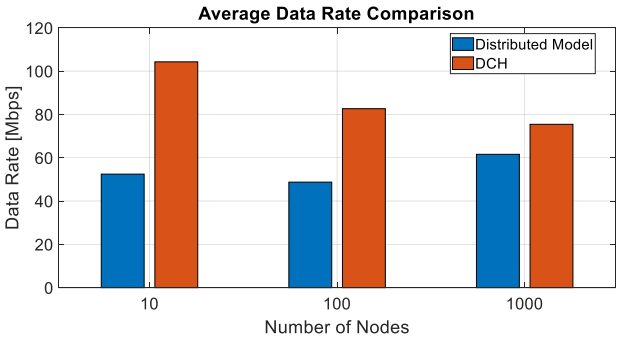


FIGURE 20. Data rate comparison between the proposed DCH and the distributed model.

data rate difference diminishes, with both models achieving nearly identical performance. These results indicate that the DCH model is better suited for scenarios requiring high data throughput, particularly in small or medium sizes networks.

V. LIMITATIONS DISCUSSION

While the proposed Dynamic Connectivity Hub (DCH) provides an innovative and efficient solution for optimizing IoT connectivity, certain limitations and potential challenges must be acknowledged to offer a balanced view of its applicability in real-world scenarios. These limitations include varying ISP performance, dependency on real-time data accuracy, and the operational challenges associated with maintaining network reliability.

A. VARYING ISP PERFORMANCE

The DCH relies on real-time metrics collected from multiple ISPs to optimize traffic distribution. However, ISP performance can fluctuate due to factors such as:

- **Congestion during peak usage hours:** ISPs may experience reduced data rates and increased latency under heavy traffic conditions.
- **Unpredictable outages:** Temporary failures or disruptions in ISP services can affect the availability of optimized routes.

These variations can challenge the DCH's ability to provide consistent Quality of Service (QoS) without frequent reconfiguration or updates. In extreme cases, a lack of reliable ISPs in a given region may limit the effectiveness of the DCH.

B. IMPACT OF NETWORK MAINTENANCE

ISP networks require periodic maintenance, which can result in temporary downtimes or degraded performance. During these events, the DCH must adapt quickly to reroute traffic to alternative ISPs, potentially leading to increased switching overheads. Prolonged maintenance windows may significantly reduce the number of available ISPs, thereby affecting the hub's ability to balance traffic effectively.

C. DEPENDENCE ON REAL-TIME DATA ACCURACY

The DCH's optimization framework, particularly its Artificial Neural Network (ANN), relies heavily on the accuracy and timeliness of the collected metrics (e.g., data rate, latency, congestion). Inaccurate or delayed data can lead to suboptimal traffic allocation, reduced QoS, and higher operational costs. For instance:

- **Latency underestimation** might result in selecting an ISP unsuitable for time-sensitive traffic.
- **Cost miscalculation** may cause over-reliance on expensive ISPs, reducing cost efficiency.

D. SCALABILITY IN LARGE-SCALE DEPLOYMENTS

In large-scale IoT networks, ensuring connectivity across distributed nodes is critical for reliable operation. While the DCH framework is designed to support multiple ISPs and diverse IoT devices, scalability in densely populated urban environments or large-scale industrial IoT deployments poses a challenge. Managing a high volume of connections, data streams, and ISP options may lead to increased computational overhead and latency in decision-making, especially during peak loads.

Distributed estimation methods, such as those proposed by Yinyan Zhang et al. [28], provide an efficient mechanism for testing and maintaining connectivity in distributed systems. By leveraging algebraic connectivity, these methods enable decentralized verification of network robustness, which aligns well with the DCH's decentralized optimization approach. Incorporating such distributed methods into the DCH framework could enhance its scalability and reliability, particularly in environments with dynamic or rapidly evolving connectivity requirements.

E. ENERGY REQUIREMENTS IN OFF-GRID SCENARIOS

Although the DCH incorporates renewable energy solutions, its reliance on battery backups and solar power can be a limitation in regions with insufficient sunlight or during prolonged power outages. Ensuring sustained operation in such environments may require additional investments in energy storage or hybrid power systems, which could increase deployment costs.

F. SECURITY CONCERNS

The multi-ISP aggregation model introduces potential security risks, as data transmitted across different networks may be susceptible to interception or breaches. While the DCH can incorporate encryption and secure routing mechanisms, these features may not fully eliminate risks associated with compromised ISP networks.

G. FUTURE WORK

To address these limitations, several mitigation strategies are proposed and can be considered such as:

- **Predictive Algorithms:** Incorporate machine learning models that predict ISP performance trends to proactively adapt to varying conditions.
- **ISP Redundancy:** Establish connections with backup ISPs to mitigate the impact of outages or maintenance.
- **Decentralized Decision-Making:** Explore distributed optimization methods to reduce computational bottlenecks in large-scale deployments.
- **Energy Optimization:** Develop hybrid power solutions combining renewable sources with grid-based or auxiliary power options.
- **Enhanced Security Layers:** Introduce advanced encryption protocols and secure data routing to mitigate ISP-related security risks.
- **Expansion** of the model to consider heterogeneous IoT networks and devices.
- **Incorporation** of proactive traffic prediction models for real-time load balancing.
- **Integration** of distributed estimation methods to add a future enhancement for the DCH in large-scale IoT networks.

Acknowledging these limitations provides a comprehensive perspective on the DCH's capabilities and sets the stage for future advancements to overcome these challenges.

VI. CONCLUSION

In this paper, we have proposed the Dynamic Connectivity Hub, DCH, as a significant innovation in multi-ISP aggregation and traffic optimization for IoT environments. We consider DCH as a generalization for our previous models that were limited to specific types of ISPs, such as LEO-GEO satellite systems. By incorporating terrestrial, mobile, and satellite ISPs, the DCH offers a comprehensive solution capable of optimizing network performance in real-time, ensuring that IoT devices remain connected to the most suitable ISP based on a variety of factors, including data rate, latency, cost, and congestion.

At the core of the DCH is an ANN that continuously analyzes input metrics from connected ISPs and dynamically adjusts the data rate allocation. This ANN-based load balancing ensures that traffic is optimally distributed across available ISPs, with sub-optimal ISPs being discarded when necessary to avoid performance degradation. Therefore, the DCH achieves significant improvements in network

resilience, latency reduction, and cost efficiency compared to traditional single-ISP and static ISP routing models. The system also demonstrates a clear advantage in load management, especially in scenarios where ISP performance fluctuates due to congestion or network outages.

One of the key strengths of the DCH is its ability to adapt to different geographic environments. In urban areas, the DCH fills connectivity gaps, such as geographical dead zones, or acts as a backup system in cases of ISP failures or overloads. In rural and remote areas, where ISPs may lack infrastructure or be unwilling to invest, the DCH serves as the primary coverage provider, utilizing satellite and mobile ISPs to deliver reliable connectivity. This flexibility makes the DCH suitable for diverse use cases, from smart cities to off-grid IoT applications in agriculture, energy, and disaster recovery.

Furthermore, the energy efficiency of the DCH contributes to its operational sustainability. Each unit is designed to run on renewable energy sources, primarily solar power, with battery backup. This ensures continuous operation, even in areas with limited or unstable access to the power grid, enhancing the system's resilience during power outages. The emphasis on energy-efficient designs makes the DCH well-suited for long-term deployments in remote and harsh environments, where access to both reliable connectivity and power is limited.

The simulation results presented in this paper demonstrate that the DCH outperforms traditional models across several key metrics. The latency reduction achieved by dynamically routing traffic through the lowest-latency ISPs is a significant benefit in time-sensitive applications such as real-time monitoring and voice communication. Simulations demonstrate that the DCH is capable of reducing latency by up to 50% compared to traditional static models, while optimizing data rates and lowering operational costs by 20-25%. The cost-saving potential of the DCH is evident in its ability to select low-cost ISPs during non-critical periods, offering businesses and organizations a flexible solution to manage their operational expenses without compromising on performance. These enhancements make the DCH particularly well-suited for urban areas with high traffic loads and rural environments where ISP infrastructure is limited.

Looking forward, the DCH model can be further enhanced by incorporating advanced features such as AI-driven traffic prediction and proactive ISP selection. With the rapid evolution of technologies like 6G, the DCH could be integrated with next-generation ISPs to provide even greater levels of performance, including ultra-low latency, higher bandwidth, and improved mobility. Moreover, the security of DCH systems presents an important avenue for future research. While the current model focuses on optimizing connectivity, future iterations could incorporate secured ISP layers for sensitive applications, such as government, military, and financial services, ensuring that critical data is transmitted over dedicated secure networks. Overall, the DCH represents a scalable, sustainable, and high-performance solution to the growing demands for reliable connectivity in an increasingly heterogeneous and data-driven world.

REFERENCES

[1] A. Aijaz and M. Sooriyabandara, "The tactile Internet for industries: A review," *Proc. IEEE*, vol. 107, no. 2, pp. 414–435, Feb. 2019.

[2] I. F. Akyildiz, "Wireless sensor networks: A survey," *Comput. Netw.*, vol. 38, no. 4, pp. 393–422, 2002.

[3] N. Z. Abdullah, I. Ahmad, and I. Hussain, "A survey of routing algorithms in software-defined networks," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 1, pp. 464–486, 1st Quart., 2018.

[4] F. Bonomi, R. Milito, P. Natarajan, and J. Zhu, "Fog computing: A platform for Internet of Things and analytics," in *Big Data and Internet of Things: A Roadmap for Smart Environments*. Cham, Switzerland: Springer, 2014, pp. 169–186.

[5] L. Li, K. Xu, T. Li, K. Zheng, C. Peng, D. Wang, and R. Mijumbi, "Multi-path TCP with multiple cellular paths: A measurement study," in *Proc. Conf. ACM Special Interest Group Data Commun.*, 2018, pp. 161–175.

[6] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 4, pp. 2322–2358, 4th Quart., 2017.

[7] N. McKeown, T. Anderson, H. Balakrishnan, G. Parulkar, L. Peterson, J. Rexford, S. Shenker, and J. Turner, "OpenFlow: Enabling innovation in campus networks," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 38, no. 2, pp. 69–74, Mar. 2008.

[8] U. L. Khan, I. Yaqoob, N. H. Tran, S. M. A. Kazmi, T. N. Dang, and C. S. Hong, "Edge computing for 5G-enabled IoT: A comprehensive survey," *IEEE Internet Things J.*, vol.7, no. 10, pp. 10200–10232, 2020.

[9] A. Beloglazov, R. Buyya, Y. C. Lee, and A. Zomaya, "Energy-efficient data centers: A survey, taxonomy, and open challenges," *Adv. Comput.*, vol. 82, pp. 47–111, 2011.

[10] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*. Cambridge, MA, USA: MIT Press, 2018.

[11] J. Kaivo-Oja, P. Virtanen, H. Jalonen, and J. Stenvall, "Big data and knowledge management in IoT systems: A review," in *Proc. 10th Int. Conf. Knowledge Manage.* Maribor, Slovenia: Springer, 2015.

[12] P. Kamboj, S. Pal, S. Bera, and S. Misra, "Traffic-aware multi-path load balancing scheme for SDN-based IoT networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 2, pp. 723–732, 2022.

[13] Z. Qin, "SD-WAN and 5G: Better together," in *Proc. SHS Web Conf., EDP Sci.*, vol. 144, 2022, pp. 1–6.

[14] S. Deng, H. Zhao, W. Fang, J. Yin, S. Dustdar, and A. Y. Zomaya, "Edge intelligence: The confluence of edge computing and artificial intelligence," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7457–7469, Aug. 2020.

[15] A. Saboor, A. Lekidis, F. Paci, and N. Zannone, "Hybrid access control for IoT networks with multiple radio interfaces," *IEEE Sensors J.*, vol. 19, no. 19, pp. 8543–8555, 2018.

[16] T. Trost and W. Granzer, "Evaluation of cost-effective multi-ISP load balancing strategies," *IEEE Netw.*, vol. 32, no. 6, pp. 102–108, Jun. 2018.

[17] M. M. Saeed, "Security and network approach in smart environments-role of SD-WAN technology," *Res. Square*, vol. 1, pp. 1–17, Mar. 2023.

[18] G. Rzym, Z. Duliński, and P. Chołda, "Dynamic link metric selection for traffic aggregation and multipath transmission in software-defined networks," *Appl. Sci.*, vol. 14, no. 12, p. 5312, Jun. 2024.

[19] L. Wang, J. Tao, and R. Ranjan, "Network traffic prediction and intelligent traffic scheduling in software defined networks," *IEEE Trans. Netw. Service Manag.*, vol. 15, no. 4, pp. 1496–1505, Apr. 2018.

[20] N. Demaku and A. Dermaku, "Improving load balancing and scalability by implementing path selection on BGP using multi SD-WAN," *J. Commun.*, vol. 17, no. 4, pp. 250–259, 2022.

[21] A. Seid, "Telecom engineering and network optimization for IoT," Ph.D. thesis, Dept. Elect. Comput. Eng., Addis Ababa Univ., Addis Ababa, Ethiopia, 2019.

[22] S. Yang, R. Chen, L. Cui, and X. Chang, "Intelligent segment routing: Toward load balancing with limited control overheads," *Big Data Mining Analytics*, vol. 6, no. 1, pp. 55–71, Mar. 2023.

[23] Y. Zhang and T. Huang, "SD-WAN for next-generation IoT networks: State-of-the-art and perspectives," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 3, pp. 1594–1626, 3rd Quart., 2020.

[24] S. Manzoor, F. Mazhar, A. Binaris, M. U. Hassan, F. Rasab, and H. G. Mohamed, "An adaptive symmetrical load balancing scheme for next generation wireless networks," *Symmetry*, vol. 15, no. 7, p. 1316, Jun. 2023.

[25] S. Manzoor, M. A. Kayani, N. Ali, N. I. Ratyal, and H. G. Mohamed, "TiWA: Achieving tetra indicator Wi-Fi associations in software defined Wi-Fi networks," *IEEE Access*, vol. 11, pp. 89520–89534, 2023.
[26] S. Manzoor, N. I. Ratyal, and H. G. Mohamed, "Achieving QoS in smart cities using software defined Wi-Fi networks," *IEEE Access*, vol. 11, pp. 98256–98268, 2023.
[27] S. Manzoor, Z. Chen, Y. Gao, X. Hei, and W. Cheng, "Towards QoS-aware load balancing for high density software defined Wi-Fi networks," *IEEE Access*, vol. 8, pp. 117623–117638, 2020.
[28] Y. Zhang, S. Li, and J. Weng, "Distributed estimation of algebraic connectivity," *IEEE Trans. Cybern.*, vol. 52, no. 5, pp. 3047–3056, May 2022.
[29] H. Ali, M. Abouelatta, and K. Y. Youssef, "On use of LEO-GEO hybrid model for optimized data traffic performance," *Wireless Pers. Commun.*, vol. 138, no. 1, pp. 641–657, Sep. 2024.
[30] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, "Internet of Things (IoT): A vision, architectural elements, and future directions," *Future Gener. Comput. Syst.*, vol. 29, no. 7, pp. 1645–1660, Sep. 2013.
[31] A. Gupta and R. K. Jha, "A survey of 5G network: Architecture and emerging technologies," *IEEE Access*, vol. 3, pp. 1206–1232, 2015.
[32] I. Ishaq, D. Carels, G. Teklemariam, J. Hoebeke, F. Abeele, E. Poorter, I. Moerman, and P. Demeester, "IETF standardization in the field of the Internet of Things (IoT): A survey," *J. Sensor Actuator Netw.*, vol. 2, no. 2, pp. 235–287, Apr. 2013.
[33] M. B. Ali, A. Gregory, and S. Li, "Edge computing: A review of multi-access edge computing," *IEEE Access*, vol. 9, pp. 18706–18721, 2021.
[34] V. Hassija, V. Chamola, V. Saxena, D. Jain, P. Goyal, and B. Sikdar, "A survey on IoT security: Requirements, threats, and solutions," *IEEE Access*, pp. 82721–82743, 2019.
[35] (2025). *Image Generation*. Accessed: Jan. 10, 2025. [Online]. Available: https://app.leonardo.ai/image-generation



HANY ALI was born in Cairo, Egypt. He received the degree in telecommunication engineering from Military Technical College, in 2008, and the master's degree from the Department of Electrical Engineering, Faculty of Engineering, Cairo University, in 2017. He is currently pursuing the Ph.D. degree with the Electrical Engineering Department, Faculty of Engineering, Ain Shams University, Egypt. In 2009, he joined the Egyptian space program for co-manufacturing satellites with Russia. He is the Main Designer and the Head of Technical Affairs of the modern Egyptian microwave communication systems. He works for the Egyptian Air Forces (EAF). He has published several articles in these fields, since 2015. His research interests include satellite communication and cognitive radios.



MOHAMED A. ABOUELATTA was born in Cairo, Egypt. He received the B.Sc. and M.Sc. degrees in electronics and communication engineering from Ain Shams University (ASU), in 1996 and 2001, respectively, and the Ph.D. degree from Ain Shams University and the National Institute of Applied Sciences (INSA) of Lyon, France, in 2010. He is currently an Associate Professor with the Faculty of Engineering, ASU. He also showcased his work at the "Innovations for The Future" Exhibition at Ain Shams University, in 2015. In addition, he co-authored and published a book titled *Heterogeneous Circuits Insights through Substrate Coupling: Noises and Parasites* (Nova Science Publishers Inc., Hauppauge, NY, USA, 2010), which appeared in the *International Journal of Circuits, Systems and Signal Processing*. His research interests include the modeling, designing, and characterization of power devices and nano-devices, the design of smart power integrated circuits, and photovoltaic and 3D CMOS heterogeneous integrated circuits. Throughout his career, he has received numerous accolades, including a "Distinction with Honor" degree, in 1996; the Ain Shams University Award, in March 2003; and the First Paper Award at the NRSC'2015 Conference, in 2015.



KHALED Y. YOUSSEF (Member, IEEE) received the Ph.D. degree from Bell Labs, USA, in 2010, and the M.B.A. degree from Cardiff Metropolitan University, U.K., in 2016. He is currently a Professor of space technology and the Vice Dean of Postgraduate Studies with the School of Navigation Science and Space Technology, Beni-Suef University. He has been a Postdoctoral Researcher with Massachusetts Institute of Technology (MIT), since 2017. He has 26 years of practical experience in Europe and the Middle East Region. He has more than 31 international publications in the security and network architecture domain published in IEEE and IET journals. Moreover, he supervised more than 25 theses of master's and Ph.D. student programs. He is also certified in digital transformation success management and deep technologies from MIT and energy efficiency audit programs from RCREE. He is a certified ISO 21500 Lead Auditor.